

Motion in a Non-inertial Reference Frame:

- It is always possible to express eqs of motion for a system in an inertial frame. But there are some problems for which these equations would be extremely complex, and it becomes easier to treat the motion of the system in a non-inertial frame of reference.
- To describe the motion on earth it is convenient to choose a reference frame on earth. However, earth undergoes a complicated motion, compounded of many different rotations (and hence accelerations) with respect to an inertial reference frame identified by "fixed" stars. Therefore, Earth reference frame is actually a non-inertial reference frame. However, solutions to many problems can be obtained to the ~~the~~ desired degree of accuracy.
- ^{Not,} Tidal forces (tidal waves) ~~can~~ can only be explained when earth coordinate system is considered as a non-inertial reference frame.
- In analysing the motion of rigid bodies it is also useful to use non-inertial reference frames.
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$$\left(\frac{d\vec{r}'}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} + \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{r}$$

$$\vec{v}_f \equiv \dot{\vec{r}}_f \equiv \left(\frac{d\vec{r}'}{dt}\right)_{\text{fixed}} : \text{velocity relative to fixed ones.}$$

$$\vec{V} \equiv \dot{\vec{R}}_f \equiv \left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} : \text{linear velocity of the ~~fixed~~ moving origin.}$$

$$\vec{v}_r \equiv \dot{\vec{r}}_r \equiv \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} : \text{velocity relative to rotating ones.}$$

$\vec{\omega}$: angular velocity of rotating ones.

$\vec{\omega} \times \vec{r}$: velocity due to the rotation of the moving ones.

$$\boxed{\vec{V}_f = \vec{V} + \vec{v}_r + \vec{\omega} \times \vec{r}}$$

Centrifugal and Coriolis Forces

Newton's ~~laws~~ laws are only valid in inertial reference frames

$$\vec{F} = m\vec{a}_f = m\left(\frac{d\vec{v}_f}{dt}\right)_{\text{fixed}}$$

$$\left(\frac{d\vec{v}_f}{dt}\right)_{\text{fixed}} = \underbrace{\left(\frac{d\vec{V}}{dt}\right)_{\text{fixed}}}_{\text{we will denote this term by}} + \left(\frac{d\vec{v}_r}{dt}\right)_{\text{fixed}} + \vec{\omega} \times \vec{r} + \vec{\omega} \times \left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}}$$

$$\equiv \ddot{\vec{R}}_f \equiv \left(\frac{d\vec{V}}{dt}\right)_{\text{fixed}}$$

Second term can be obtained by replacing $\vec{Q}$ with $\vec{v}_r$ in the below equation:

$$\left(\frac{d\vec{Q}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{Q}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{Q}$$

Therefore,

$$\left(\frac{d\vec{v}_r}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{v}_r}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{v}_r$$

$$\equiv \vec{a}_r + \vec{\omega} \times \vec{v}_r \quad \text{where } \vec{a}_r \text{ is the acceleration in rotating coordinate system.}$$

The last term can be straightforwardly obtained by considering:

$$\left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{r} \Rightarrow \vec{\omega} \times \left(\frac{d\vec{r}}{dt}\right)_{\text{fixed}} = \vec{\omega} \times \left(\frac{d\vec{r}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

$$= \vec{\omega} \times \vec{v}_r + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

If we combine the equations we obtained until now:

$$\vec{F} = m \vec{a}_f = m \ddot{\vec{R}}_f + m \vec{a}_r + m \vec{\omega} \times \vec{v}_r + m \vec{\omega} \times \vec{v}_r + m \vec{\omega} \times (\vec{\omega} \times \vec{r}) + m \vec{\omega} \times \vec{r}$$

$$= m \ddot{\vec{R}}_f + m \vec{a}_r + m \vec{\omega} \times \vec{r} + m \vec{\omega} \times (\vec{\omega} \times \vec{r}) + 2m \vec{\omega} \times \vec{v}_r$$

To an observer in the rotating coordinate system:

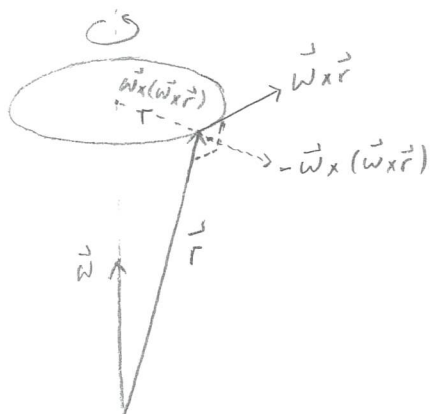
$$m \vec{a}_r = \vec{F} - m \ddot{\vec{R}}_f - m \vec{\omega} \times \vec{r} - m \vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m \vec{\omega} \times \vec{v}_r$$

[This result was first published by G. G. Coriolis in 1835 about composition of acc. for water wheels.]

$\vec{F}$: sum of the forces acting on the particle as measured in the fixed system.

$-m \ddot{\vec{R}}_f$ } Result of translational and angular accelerations ($\ddot{\vec{R}}_f$ and $\dot{\vec{\omega}}$)
 $-m \vec{\omega} \times \vec{r}$ } of the moving coordinate with respect to the fixed system.

$-m \vec{\omega} \times (\vec{\omega} \times \vec{r})$: Centrifugal force



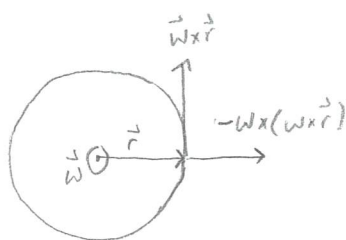
$-\vec{\omega} \times (\vec{\omega} \times \vec{r})$ points outwards, away from the axis of rotation.

~~Coriolis force~~

Note that when $\vec{\omega} \perp \vec{r}$: $-m \vec{\omega} \times (\vec{\omega} \times \vec{r}) = -m \omega^2 \vec{r}$

$$|\vec{\omega} \times \vec{r}| = |\omega| |\vec{r}|$$

$$|\omega \times (\omega \times \vec{r})| = |\omega|^2 |\vec{r}|$$



- The last term ($-2m\vec{\omega} \times \vec{v}_r$) is a new quantity and it's called "Coriolis force". It arises from the motion of particle in the rotating coordinate system.
 $\Rightarrow$ If the particle is at rest in the rotating coordinate system, Coriolis force vanishes.

★ Centrifugal and Coriolis forces are not forces in the usual sense. We introduced them in an artificial manner to write an equation resembling Newton's equation:

In an inertial reference frame:

$$\vec{F} = m\vec{a}_f$$

$\Leftrightarrow$

In an non-inertial reference frame:

$$\vec{F}_{\text{eff}} = m\vec{a}_r$$

Remember:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r}_1 + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}_1$$

$$\vec{F}_{\text{eff}} = \vec{F} - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}_r$$

$\downarrow$
 $m\vec{a}_f$

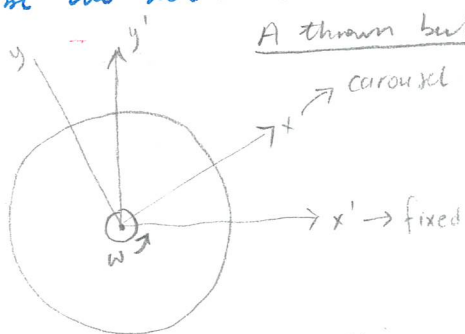
[Let's assume $\vec{R}_f$ and $\vec{\omega}$ are zero for simplicity]

$$\vec{F}_{\text{eff}} = m\vec{a}_f + (\text{noninertial terms})$$

where noninertial terms are identified as centrifugal and Coriolis forces.

- ★ For example, if a body rotates about a fixed force center the only real force is the force of attraction towards the center (and it gives rise to centripetal acceleration.) An observer moving with the rotating body, however, measures this central force and also notes that the body does not fall toward the force center. To explain this situation, with the requirement that the net force on the body vanish, the observer must postulate an additional force - the anti-fugal force. This requirement is artificial; it solely arises from the attempt to enter the form of Newton's equation to a noninertial system, and this can be done only by introducing a fictitious "correction force". Same applies for Coriolis force; it arises when an attempt is made to describe the motion relative to the rotating body.

A thrown ball example:



$$\vec{F}_{\text{eff}} = \vec{F} - m\vec{R}_f - m\vec{\omega} \times \vec{r} - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}_r$$

$\downarrow \quad \downarrow \quad \downarrow$
0 0 0

$$\vec{F}_{\text{eff}} = \underbrace{-m\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{centrifugal}} - \underbrace{2m\vec{\omega} \times \vec{v}_r}_{\text{Coriolis}}$$

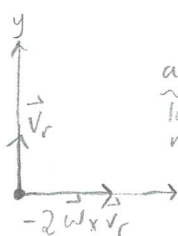
For purposes of calculation:

$\omega = 1 \text{ rad/s}$, $R = 1 \text{ m}$, ball is thrown in y direction.

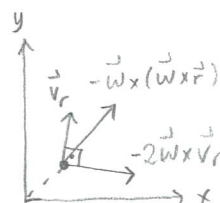
$$\vec{a}_{\text{eff}} = \vec{F}_{\text{eff}}/m = \underbrace{-\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{always directed radially outward}} - 2\vec{\omega} \times \vec{v}_r$$

$\rightarrow$ It is hard to calculate the path. But numerical simulations would help.

Therefore, at the moment the ball is thrown:



a bit later maybe



CENTRIFUGAL:

$$\vec{\omega} = \omega_0 \hat{z}, \quad \vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{\omega} \times \vec{r} = \omega_0 x \hat{j} - \omega_0 y \hat{i}$$

$$\vec{\omega} \times (\vec{\omega} \times \vec{r}) = -\omega_0^2 x \hat{i} - \omega_0^2 y \hat{j}$$

$$-\vec{\omega} \times (\vec{\omega} \times \vec{r}) = +\omega_0^2 x \hat{i} + \omega_0^2 y \hat{j}$$

CORIOLIS:

$$\vec{\omega} = \omega_0 \hat{z}, \quad \vec{v}_r = v_{rx} \hat{i} + v_{ry} \hat{j}$$

$$\vec{\omega} \times \vec{v}_r = \omega_0 v_{rx} \hat{j} - \omega_0 v_{ry} \hat{i}$$

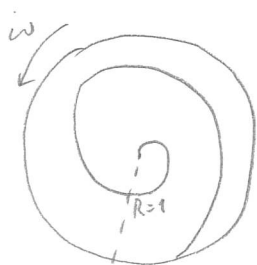
$$-2\vec{\omega} \times \vec{v}_r = 2\omega_0 v_{ry} \hat{i} - 2\omega_0 v_{rx} \hat{j}$$



TOTAL ACCELERATION:

$$-\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2\vec{\omega} \times \vec{v}_r = [\omega_0^2 x + 2\omega_0 v_{ry}] \hat{i} + [-2\omega_0 v_{rx} + \omega_0^2 y] \hat{j}$$

... see the Python code for numerical simulation



$$\omega = 1 \frac{\text{rad}}{\text{s}} \text{ counter clockwise}$$

$$\vec{v}_r = 1 \hat{j} \frac{\text{m}}{\text{s}}$$

* From the point of view of x' coordinates, the ball simply goes straight in y -direction.

• Once we know $\vec{a}_{\text{eff}} = \frac{\vec{F}_{\text{eff}}}{m} = -\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2\vec{\omega} \times \vec{v}_r$

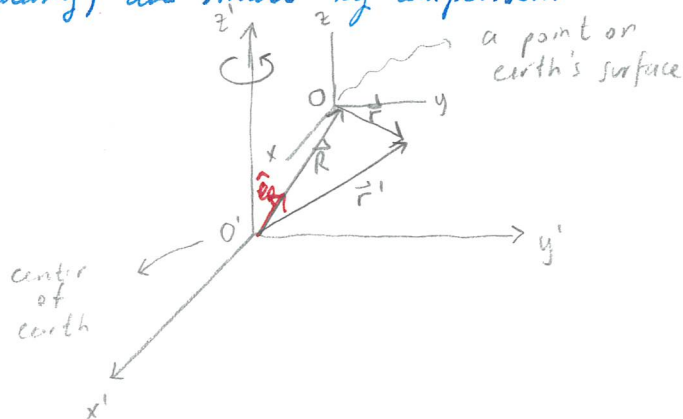
we can calculate $\vec{v}_{\text{eff}}$ and $\vec{r}_{\text{eff}}$ numerically by integration:

$$\vec{v}_{\text{eff}} = \int \vec{a}_{\text{eff}} dt$$

$$\vec{r}_{\text{eff}} = \int \vec{v}_{\text{eff}} dt$$

Motion relative to the Earth

The motion of earth w.r.t. an inertial reference frame is dominated by earth's own rotation about its own axis. The effects of other motions (e.g. the revolution about the sun and the motion of the solar system w.r.t. the local galaxy) are small by comparison.



Total force from x' frame:

~~Force~~

$$\vec{F} = \vec{S} + m \vec{g}_0 \quad \text{Earth's gravitational acc. vector.}$$

sum of external forces except the gravitational force (friction, electromagnetic, etc.)

$$\vec{g}_0 = -G \frac{M_E}{R^2} \hat{e}_R$$

M_E : mass of earth

$\hat{e}_R$: unit vector along the direction of $\vec{R}$.

NOTE: Acc. of gravity varies over the surface of Earth due to Earth's oblateness, density of nonuniformities, and altitude. But, here we neglect these effects.

$$\vec{F}_{\text{eff}} = \vec{S} + m\vec{g}_0 - m\ddot{\vec{R}}_f - m\vec{\omega} \times \vec{r} - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}_r$$

$$\vec{\omega} = 7.3 \times 10^{-5} \frac{\text{rad}}{\text{s}} \hat{e}_z \rightarrow \text{relatively slow but it is 365 times greater than the rotation frequency of the Earth about sun.}$$

$$\vec{\omega} \approx 0 \rightarrow \vec{\omega} = \text{constant (practically)} \Rightarrow \vec{\omega} \times \vec{r} \text{ will be neglected.}$$

The term $-m\ddot{\vec{R}}_f$:

According to $\left(\frac{d\vec{Q}}{dt}\right)_{\text{fixed}} = \left(\frac{d\vec{Q}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{Q}$ where $\vec{Q}$ is any vector,

$$\left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} = \ddot{\vec{R}}_{\text{fixed}} = \left(\frac{d\vec{R}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{R} = \vec{\omega} \times \vec{R}$$

If we use the same expression for $\vec{R}$ again:

$$\left(\frac{d\vec{R}}{dt}\right)_{\text{fixed}} = \dot{\vec{R}}_f = \left(\frac{d\vec{R}}{dt}\right)_{\text{rotating}} + \vec{\omega} \times \vec{R} = \vec{\omega} \times \vec{R}$$

$$\Rightarrow \ddot{\vec{R}}_f = \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

Therefore $\vec{F}_{\text{eff}}$ becomes:

$$\begin{aligned} \vec{F}_{\text{eff}} &= \vec{S} + m\vec{g}_0 - m\ddot{\vec{R}}_f - m\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2m\vec{\omega} \times \vec{v}_r \\ &= \vec{S} + m\vec{g}_0 - \underbrace{m\vec{\omega} \times [\vec{\omega} \times (\vec{r} + \vec{R})]}_{\text{centrifugal force}} - 2m\vec{\omega} \times \vec{v}_r \end{aligned}$$

★ Second and third terms are what we measure on earth as the effective $\vec{g}$:

$$\vec{g} = \vec{g}_0 - \vec{\omega} \times [\vec{\omega} \times (\vec{r} + \vec{R})]$$

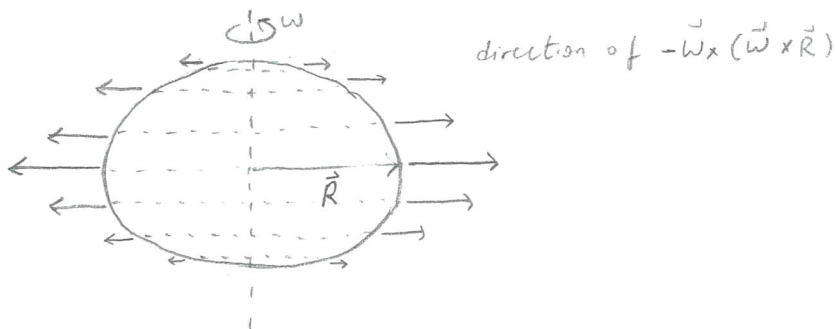
Therefore, $\vec{F}_{\text{eff}}$ can be written as:

$$\vec{F}_{\text{eff}} = \vec{S} + m\vec{g} - 2m\vec{\omega} \times \vec{v}_r$$

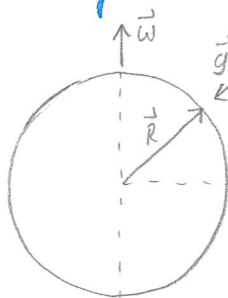
NOTE: If we are considering the motion near the surface of Earth, then $r \ll R$ and the second term is completely dominated by $\vec{\omega} \times (\vec{\omega} \times \vec{R})$

- ★ The centrifugal force responsible for the oblateness of the Earth; Earth is not a solid spheroid, it's more like a viscous liquid with a solid crust.

Because of Earth's rotation, Earth has deformed so that its equatorial radius is 21.4 km greater than its polar radius, and thus the gravitational acceleration is 0.052 m/s^2 greater at the poles than at the equator.

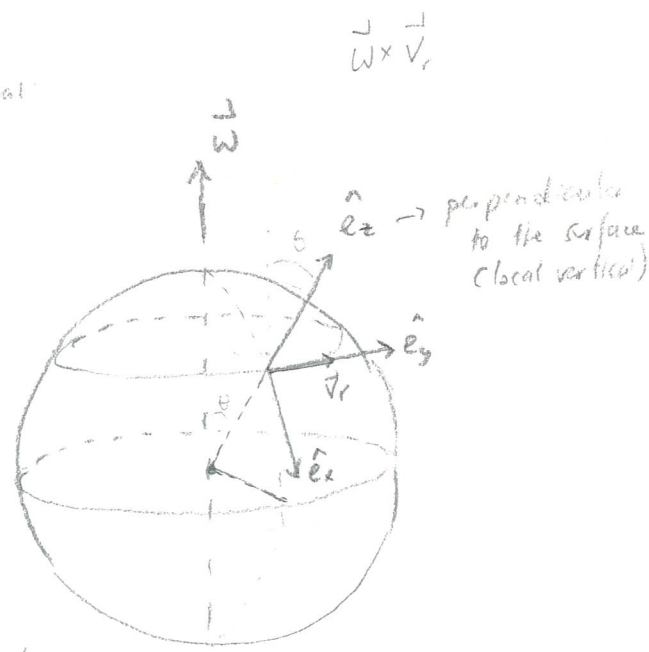
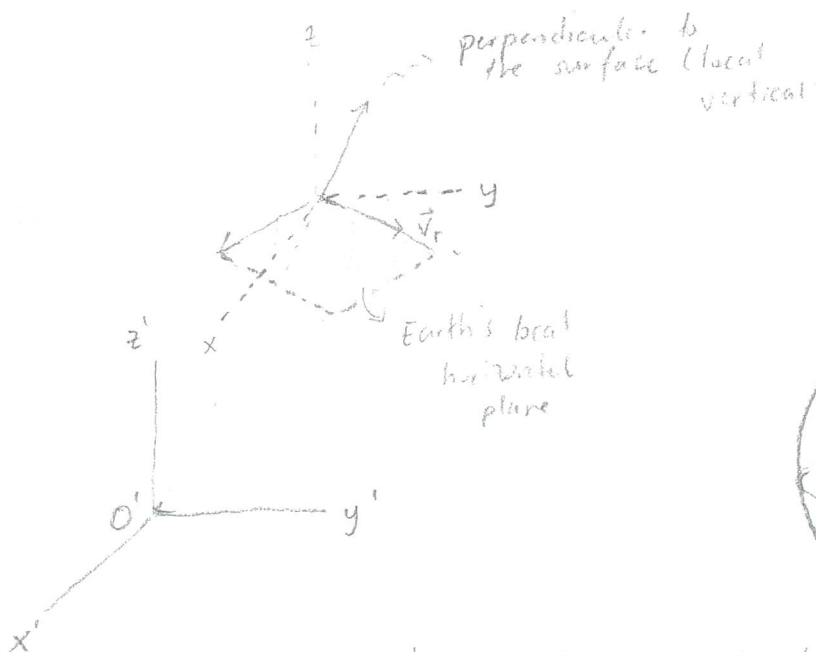


- ★ The direction of measured $\vec{g}$ according to $\vec{g} = \vec{g}_0 - \vec{\omega} \times [\vec{\omega} \times (\vec{r} + \vec{R})]$



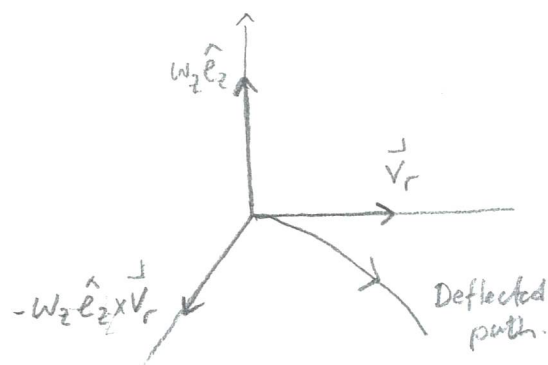
~~plumb line~~ ~~at work~~
The direction of a plumb bob will include the centrifugal term. Because of this effect, the direction of $\vec{g}$ at a given point is in general slightly different from the true vertical (defined as the direction of the line connecting the point with the centre of earth.)

Coriolis Force Effects



The angular velocity vector $\vec{\omega}$ is directed in north direction. Therefore in northern hemisphere $\vec{\omega}$ has a component ω_z directed outward along the local vertical. If the particle is projected in a horizontal plane (in local coordinates it is the surface of Earth) with a velocity $\vec{v}_r$, then the Coriolis force $-2m\vec{\omega} \times \vec{v}_r$ has a component in that plane of $2m\omega_z v_r$ directed toward the right of the particle's motion.

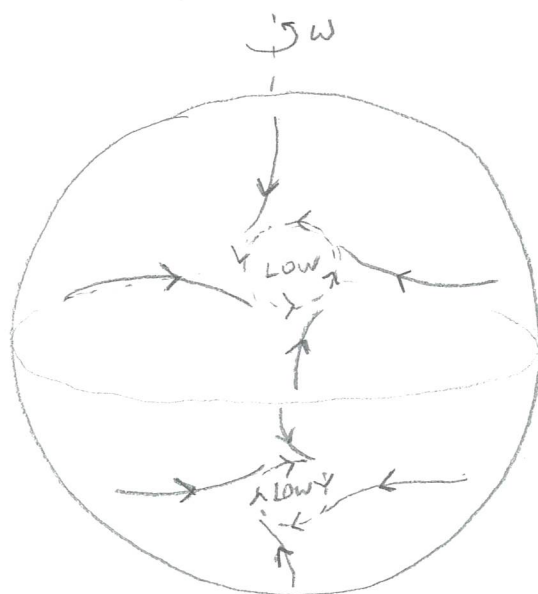
local vertical



★ Because of the magnitude of the horizontal component of the Coriolis force is proportional to the vertical component of $\vec{\omega}$, this force is max at the North pole and zero at the equator.

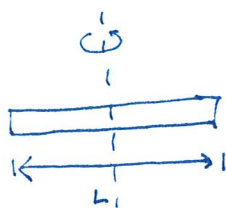
★ In Southern hemisphere the component of ω_z is directed inward along the local vertical, and hence the deflections are opposite sense from those in the Northern hemisphere.

★ The most noticeable effect of Coriolis force is that on the air masses. As air flows from high-pressure regions to low pressure, the Coriolis force deflects the air toward the right in the Northern hemisphere, producing cyclonic motion.



rotational Calculating inertias (Intro for Dynamics of Rigid bodies)

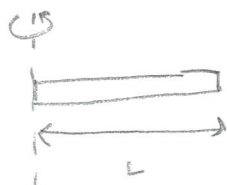
Ex: Rod about center (with ^{linear} mass distribution λ , and mass M)



$$dm = \lambda dx$$

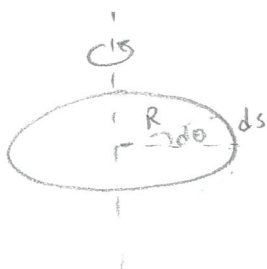
$$I = \int_{-L/2}^{+L/2} \lambda x^2 dx = \lambda \left[\frac{L^3}{8 \cdot 3} + \frac{L^3}{8 \cdot 3} \right] = \lambda \frac{2L^3}{8 \cdot 3} = \frac{1}{12} \lambda L^3 = \frac{1}{12} \frac{M}{L} L^3 = \frac{1}{12} M L^2$$

Ex: Rod about end (λ)



$$I = \int_0^L \lambda x^2 dx = \frac{1}{3} \lambda L^3 = \frac{1}{3} \frac{M}{L} L^3 = \frac{1}{3} M L^2$$

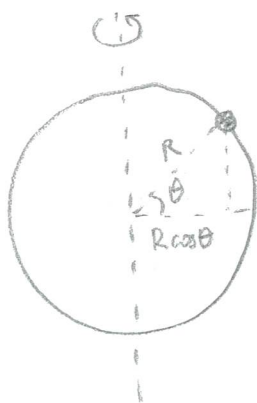
Ex: Hoop about symmetry axis (λ)



$$dm = \lambda ds$$

$$I = \int_0^{2\pi} R d\theta \lambda R^2 = R^3 \lambda \int_0^{2\pi} d\theta = R^3 \lambda 2\pi = 2\pi \frac{M}{2\pi R} R^3 = M R^2$$

Ex: Hoop about diameter (λ)



$$dm = \lambda R d\theta$$

$$I = \int_0^{2\pi} R^2 \cos^2 \theta \lambda R d\theta = \lambda R^3 \int_0^{2\pi} \cos^2 \theta d\theta$$

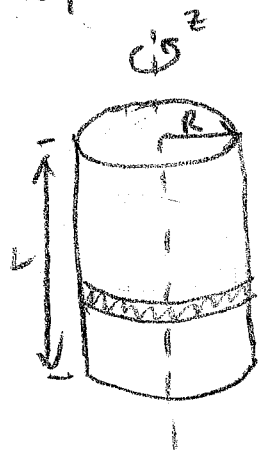
$$= \lambda R^3 \int_0^{2\pi} \left[\frac{1 + \cos(2\theta)}{2} \right] d\theta$$

$$= \lambda R^3 \left\{ \frac{1}{2} \int_0^{2\pi} d\theta + \int_0^{2\pi} \cos(2\theta) d\theta \right\}$$

$$= \lambda R^3 \left\{ \frac{1}{2} 2\pi + \frac{1}{2} \sin(2\theta) \Big|_0^{2\pi} \right\}$$

$$= \lambda R^3 \{ \pi \} = \pi \lambda R^3 = \frac{\pi}{2\pi R} \frac{M}{\pi} R^3 = \frac{1}{2} M R^2$$

Ex: Cylindrical shell about the symmetry axis (σ)



$$dA = 2\pi R dz$$

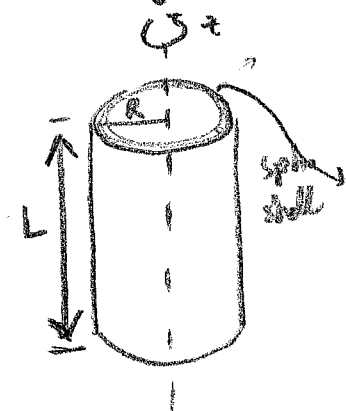
$$dm = \sigma dA = \sigma 2\pi R dz$$

$$dI = dm R^2 = \sigma 2\pi R^3 dz$$

$$I = \int dI = \int_0^L 2\pi R^3 \sigma dz = 2\pi R^3 \sigma L = \cancel{2\pi R^3} \frac{M}{\cancel{2\pi R L}} = MR^2$$

(or disk)

Ex: Solid cylinder, about symmetry axis: (g)



$$dV = 2\pi r L dr$$

$$dm = 2\pi r L g dr$$

$$dI = dm r^2 = 2\pi L g r^3 dr$$

$$\int_0^R dI = 2\pi L g \int_0^R r^3 dr = 2\pi L g \frac{R^4}{4}$$

$$= \cancel{2\pi L} \frac{M}{\cancel{2\pi L}} \frac{R^4}{4} = \frac{1}{2} MR^2$$

Same with the disc (symmetry axis)